

Synthesis Post-Mortem: Why the 3-Domain Mixed-Process Architecture Failed

1. Introduction

During the initial development of the Multi-Voltage Crypto Core, the architecture attempted to utilize three distinct voltage domains: 1.16V (High Power), 0.95V (Top Level), and 0.75V (Low Power).

However, synthesis repeatedly failed with LIBSETUP-024 (Conflicting Operating Conditions) and OPT-101 (Missing Inverter) errors. This document explains the underlying physical and tool-based constraints that made the original architecture impossible to synthesize using the provided SAED32 educational PDK.

2. Root Cause 1: The Process Corner Conflict (LIBSETUP-024)

Understanding PVT (Process, Voltage, Temperature)

In physical design, standard cell libraries are characterized under specific PVT "corners":

- ss (Slow-Slow): Assumes worst-case manufacturing (Process = 1.01). Transistors switch slower.
- ff (Fast-Fast): Assumes best-case manufacturing (Process = 0.99). Transistors switch faster.

The Physics Limitation

A single piece of physical silicon cannot be manufactured as "Slow-Slow" on one half and "Fast-Fast" on the other half. It is a single wafer.

In the original script, we attempted to map the 1.16V domain using a Fast-Fast library (saed32rvt_ff1p16v125c.db) while mapping the 0.75V domain using a Slow-Slow library (saed32rvt_ss0p75v125c.db). Design Compiler's timing engine immediately threw the LIBSETUP-024 error because it refuses to perform timing analysis across conflicting physical realities.

3. Root Cause 2: The Level Shifter Paradox (OPT-101)

To bypass the LIBSETUP-024 conflict, we attempted to use `set_target_library_subset` to lock the ff and ss libraries into strictly isolated rooms. This successfully hid the conflict from the global timing engine, but it created a fatal issue at the domain boundaries.

The Missing Bridge

To pass a signal from a 0.95V ss domain to a 1.16V ff domain, the UPF engine must insert a Level Shifter. However, a Level Shifter sits on the boundary and must operate in both domains simultaneously.

Because we isolated the subsets, the Level Shifter could not mathematically exist in both a Process=1.01 (ss) and a Process=0.99 (ff) reality at the same time. The subset crashed, dropping the library entirely. When Design Compiler tried to map the base logic for that module, it had zero libraries loaded, resulting in the OPT-101: An inverter is required for mapping crash.

4. The PDK Limitation

Why couldn't we just use a Slow-Slow library for the 1.16V domain to match the rest of the chip?

The files did not exist. The SAED32 educational PDK provided for this project was incomplete. It lacked:

1. An ss (Slow-Slow) library for 1.16V.
2. An ff (Fast-Fast) library for 0.75V.

Without a unified process corner across all three voltages, it is physically impossible for Design Compiler to stitch the domains together and perform standard cell mapping.

5. The Resolution: The 2-Domain Architecture

The solution was to drop the 1.16V requirement and build a 2-Voltage architecture (0.95V and 0.75V).

Because the PDK contained both saed32rvt_ss0p95v125c.db and saed32rvt_ss0p75v125c.db, the entire chip could be unified under the Slow-Slow (ss) process corner.

- PVT conflicts (LIBSETUP-024) were eliminated.
- Level Shifters (saed32rvt_ulvl_ss0p95v125c_i0p75v) matched the process corner perfectly.
- Subsets were no longer required, allowing Design Compiler to access all logic cells seamlessly, completely resolving the OPT-101 crashes.